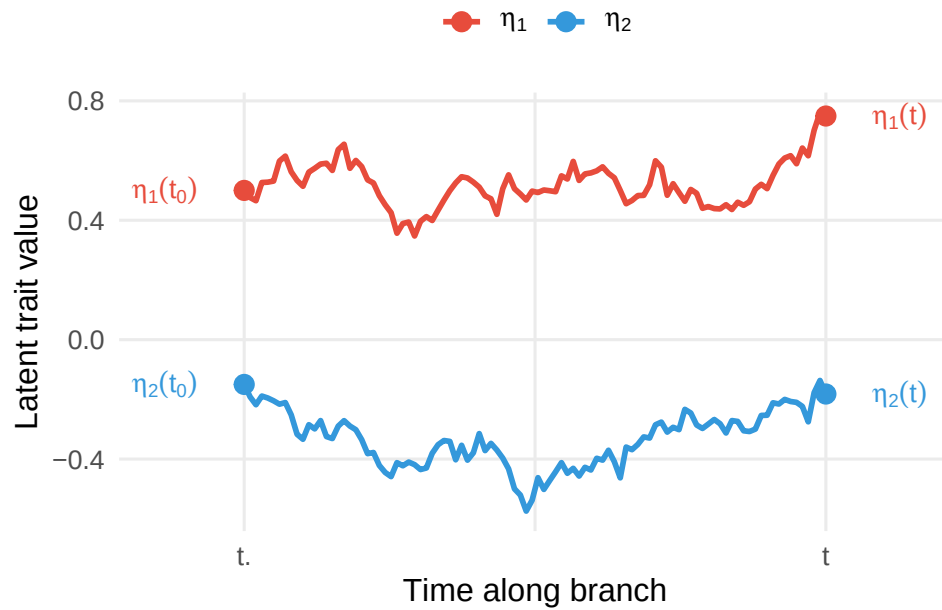


GDPM Tree Traversal Algorithm

Forward-time simulation of latent trait evolution along phylogenetic branches

Continuous trait evolution along a single branch



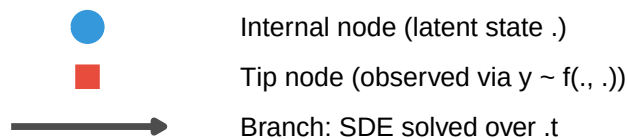
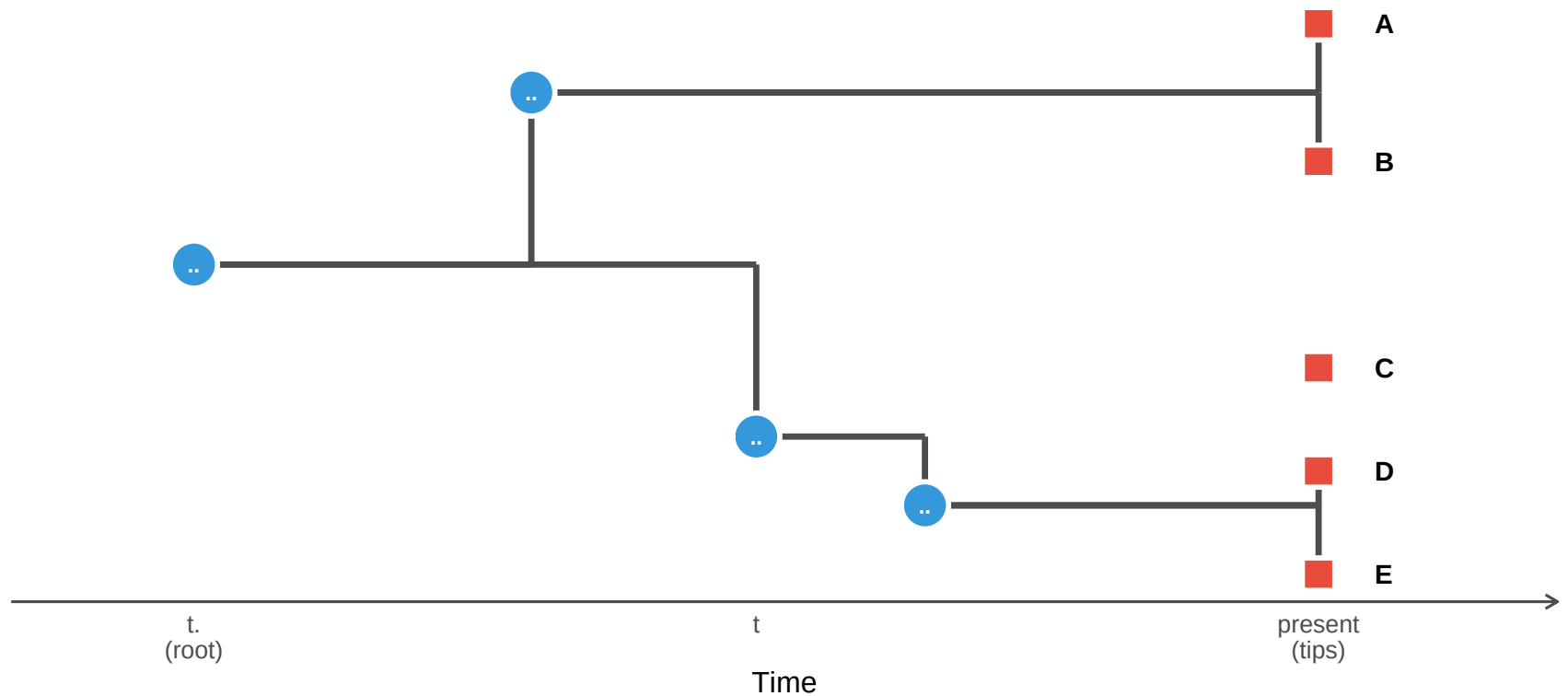
SDE Solution (per branch)

$$\boldsymbol{\eta}(t) = \mathbf{A}[\cdot] \cdot \boldsymbol{\eta}(t_0) + \mathbf{b}[\cdot] + \boldsymbol{\epsilon}$$

$$\boldsymbol{\epsilon} \sim N(\mathbf{0}, \mathbf{Q}[\cdot])$$

$$\mathbf{A}[\cdot] = \exp(\mathbf{A} \cdot \Delta t)$$

$$\mathbf{Q}[\cdot] = \mathbf{Q}[\cdot] - \mathbf{A}[\cdot] \cdot \mathbf{Q}[\cdot] \cdot \mathbf{A}[\cdot]'$$



Latent traits $\boldsymbol{\eta}$ evolve according to the SDE, with values at each node serving as initial conditions for descendant branches.
At tips, latent values are linked to observed data through the measurement model.